

Third Year Labs - Ising Model

Table of contents

1	Third Year Labs - Ising Model	4
1.1	Project Overview	4
1.2	Key Definitions Used in The Project	4
1.3	Literature Review	4
2	Log Overview	5
2.0.1	Overview of the Weekly Log	5
2.1	Week One:	5
2.2	Week Two:	5
2.3	Week Three:	5
2.4	Week Four:	5
2.5	Week Five:	6
2.6	Week Six:	6
2.7	Week Seven & Eight:	6
3	Week 1 Overview	7
4	Week 2 Overview	9
5	Week 3 Overview	11
6	Week 4 Overview	13
7	Week 5 Overview	14
8	Week 6 Overview	16
9	Week 7 & 8 Overview	18
10	Equations	19
10.1	Inverse temperature	19
10.2	Spin variables	19
10.3	Total number of sites	19
10.4	Ising Hamiltonian	19
10.5	Energy change for a trial spin flip	19
10.6	Metropolis acceptance rule	20

10.7 Periodic boundary condition	20
10.8 Magnetisation	20
10.9 Energy of a configuration	20
10.10 Cluster size	20
10.11 Cluster number density	20
10.12 Mean cluster number density	20
10.13 Standard error of the mean	21
10.14 Power-law cluster size distribution	21
10.15 Radius of gyration squared	21
10.16 Circular mean mapping	21
10.17 Cluster extent	21
10.18 Spanning criterion	21
10.19 Cyclomatic number	21
10.20 Compactness measure	22
10.21 Largest-cluster fraction	22
10.22 Autocorrelation function	22
10.23 Decorrelation time	22
10.24 Logarithmic binning	22

1 Third Year Labs - Ising Model

1.1 Project Overview

This project involves identifying clusters of like spins and measuring how their size, shape, and evolution depend on temperature, revealing how cluster geometry underlies critical behaviour in the 2D Ising model.

1.2 Key Definitions Used in The Project

- In this project, a cluster is defined as a connected component of nearest-neighbour spins with the same orientation on the lattice. Connectivity is determined purely by geometric adjacency, meaning that any neighbouring spins with the same sign belong to the same cluster. This definition matches the approach used in studies of cluster geometry such as Domb and Stoll, and allows direct measurement of cluster size, shape, and connectivity from simulation data. The physical significance of this definition is supported by the Fortuin–Kasteleyn representation of the Ising model, which shows that spin correlations correspond to probabilities of sites belonging to the same cluster, ensuring that these geometric clusters are directly linked to the underlying statistical physics of the system.

1.3 Literature Review

A literature review was conducted to understand other project conducted within the Clusters remit. Details of this can be found [here](#).

2 Log Overview

2.0.1 Overview of the Weekly Log

2.1 Week One:

- Discussed options for project direction
- Set up GitHub
- Set up Quarto Website
- Found related literature
- Read literature and performed literature review
- Identified possible direction for project

2.2 Week Two:

- Installed Spyder and Reimagined Quarto
- Edited and developed the base code
- Created code for the Clustering Algorithm, specifically in 2D
- Established Secure definition of a cluster

2.3 Week Three:

- Adapted the Clustering Algorithm code created in week 2 to extend it to 1D and 3D
- Produced a series of results and outputs for the 2D set-up
- Updated the Lab Book so that it was functional on Quarto.

2.4 Week Four:

- Wrote functional, sophisticated and transparent plots for 2D clustering, based on the 2D work from week 3.

- Began to code more sophisticated and functioning code for the 3D clustering, in-line with the literature.
- Updated Lab Book to reflect current progress

2.5 Week Five:

- Finalised the code the 3D, producing 5 relevant and functional graphs.
- Decided on a specific narrative for the project, selecting on the optimal graphs to reflect this.
- Adapted said optimal graphs to display the data for multiple temperatures around the critical value.
- Updated Lab Book to reflect current progress, as well as the equations sheet.

2.6 Week Six:

- Adapted the 2D code to aid running efficiency, ensuring large cluster sizes could be sustainably achieved.
- Continued writing the report, focusing on the theory, methods, results and references sections.
- Updated Lab Book to reflect to ensure that the weekly log was complete and the main page was detailed, ready for submission.

2.7 Week Seven & Eight:

- Worked on and Completed the report, ensuring that the most important information was included and that the most meaningful equations were present, as well as that references were cited accurately.
- Finalised the Lab Book to ensure that it was all in order for submission.

3 Week 1 Overview

3.0.0.1 Main Activities:

- Deciding direction of the project and possible avenues to explore.
 - The current top suggests are something to do with how clustering attributes (such as shapes and sizes) vary with dimensionality, or something to do with the critical exponent.
 - We are also looking into where there might be a cross over between the critical exponent and clustering attributes, and whether this could be incorporated into the project.
- Setting up infrastructure to facilitate the project
 - Set up a GitHub account so that I could pull the start-up code repository
 - Installed Quarto, found I didn't need to install Quarto and then set up Quarto access to work through Visual Studio Editor. VS Editor had to be deleted and re-installed, following which Quarto functioned.
 - I then set up a few pages on the website to become familiar with how it works.
- Literature Inquiry
 - Searched online for articles on Cluster Algorithms and the Critical Exponent. I found several articles that were potentially of interested, including one that looked like it may contain helpful coding hints.
 - I ensured that all of the article were proper science articles from reputable sources.
- Literature Review
 - I then proceeded to read the articles and write a (relatively!) brief literature review for each one.
 - One of the key observations from this was the way in which a Cluster is defined. I also established important equations and identified recommended approaches.
 - I then wrote a final literature summary and identified a possible project direction based on the content of the articles.

3.0.0.2 Main Discoveries:

- Possible Project direction considered at this point: As the spatial dimension increases, cluster formation in the Ising model changes qualitatively. In one dimension, clusters remain finite and their size distribution is exactly known, reflecting the absence of a phase transition. In higher dimensions, clusters grow, change shape, and become increasingly compact as the system approaches the critical temperature. These geometric changes are directly linked to diverging correlation lengths and critical scaling. Cluster-based representations and algorithms reveal that clusters are the fundamental objects governing both static and dynamic critical behaviour, explaining why cluster size, shape, and formation times depend sensitively on dimension.

3.0.0.3 Main Documentation:

- Please [click here](#) to see the full literature review

4 Week 2 Overview

4.0.0.1 Main Activities:

- Installed Spyder
 - Installed Spyder as a more effective IDE that was very compatible with GitHub.
- Reimagined Quarto
 - Used terminal and quarto preview to facilitate the lab book, rather than running through VS Editor.
- Edited and Developed Base Code
 - Edited and developed the code that was provided at the beginning of the project so that it was tailored to the project direction.
- Created Code for the Clustering Algorithms
 - I created the core of the code for the clustering algorithm.
 - As a starting point, I decided to focus on the cluster size, shape and speed to converge, when very close to the critical temperature.
 - I am endeavouring, at this point, to try and establish the clustering attribute outcomes for 1D, 2D and 3D beginning with the 2D code.
 - I decided that it would be better to have a set code block for 1D, 2D and 3D, rather than a set code block for the attributes and then re-coding them every time for the dimension.
 - I also decided that it would be beneficial to use Monte-Carlo Metropolis methods rather than Wolff or Swendsen due to the definition of a cluster.
 - It was also important to establish the exact definition for a cluster.

4.0.0.2 Main Discoveries:

- Code core written

- Definition of a cluster established: In this project, a cluster is defined as a connected component of nearest-neighbour spins with the same orientation on the lattice. Connectivity is determined purely by geometric adjacency, meaning that any neighbouring spins with the same sign belong to the same cluster. This definition matches the approach used in studies of cluster geometry such as Domb and Stoll, and allows direct measurement of cluster size, shape, and connectivity from simulation data. The physical significance of this definition is supported by the Fortuin–Kasteleyn representation of the Ising model, which shows that spin correlations correspond to probabilities of sites belonging to the same cluster, ensuring that these geometric clusters are directly linked to the underlying statistical physics of the system.

5 Week 3 Overview

5.0.0.1 Main Activities:

- Translated the Clustering Algorithm to extend to 1D and 3D.
 - Currently, the code created is functional for 2D only, so my starting point is to extend it to 1D & 3D.
 - My main consideration when doing this was how to put the inputs needed to vary with the dimensions, as well as which graphs would still produce correctly and which would need reconstructing.
 - A lot more graphs needed changing for the 3D, and I also included a graph for the Anisotropy.
 - Less needed changing for 1D - the only significant edit was how to graphically represent 1D shape.
- Started to produce results
 - I started to produce a few preliminary results, starting with 2D.
 - I began with a very small cluster size to confirm that the code was functioning correctly.
 - I then moved on to increasing the cluster size, noting that larger cluster sizes matched more closely with the literature. A drawback to larger cluster sizes is the configuration time.
 - I decided it was better to get concrete results for 2D before moving onto other dimensions.
 - To this end, I varied the temperature and cluster size to produce a series of results.
- Updated Lab Book
 - I put the information I had been collecting, such as the literature review notes, into the Lab book.
 - I also inputted my entries for weeks 1-3 that I had been storing on alternative software until now.

5.0.0.2 Main Results:

- Have functional code in 1D and 3D
- Produced a series of results and outputs for the 2D set-up
- Updated Lab Book

6 Week 4 Overview

6.0.0.1 Main Activities:

- Upgraded the 2D Code
 - Looking at the output results produced in 2D last week, I decided to improve the quality of these.
 - I once again researched into the literature to see if there was a more sophisticated way to display the data, as scatter graphs limit the clarity of the representation of the data.
 - I found that log plots showed a much more transparent representation of the data so made the relevant graphs into log plots.
 - I varied the 2D Inputs to establish that the plots were functional and working well.
- Started to work in 3D
 - Given the success (hopefully) of the 2D coding, I decided to proceed on to the 3D code.
 - My plan is to try the 3D coding and depending on it's success, structure my 2D inputs accordingly.
 - I began to construct the code, again reading the literature to ensure that plots were as sophisticated and transparent as possible.
 - The biggest issue for this has been the complication times, as I needed to use big enough clusters to establish whether the plots were of a good quality.

6.0.0.2 Main Results:

- Have functional, more sophisticated and transparent plots for 2D.
- More sophisticated and functioning plots for 3D in the making.
- Updated Lab Book

7 Week 5 Overview

7.0.0.1 Main Activities:

- Finished Creating the Code in 3D
 - I established which graphs would work in a direct transfer to 3D and which would need a lot of alteration / need another way to visualise the data.
 - I wrote the re-worked the code that could go from 2D to 3D smoothly and wrote the code for the graphs that needed visualising in a different way.
 - I structured the code in such a way that it would be very easy to alter the cluster size / number of steps used the in the calculation.
 - The final result for the coding was 5 comprehensive, relevant and functional graphs.
- Re-Stylised the 2D Code
 - Considering the report, I decided to specifically select which, out of the 7 plots, would be best suited to include, and hence which 4 or 5 I should concentrate on polishing.
 - Following this, I considered the narrative of the project and exactly what the graphs should convey.
 - To this end, I decided to consider 5 different temperatures of the same cluster size, and plot the results for all of these on the same graph for easy comparison.
 - I then proceeded to re-work the code to achieve this.
 - Considering the compilation time of 300x300 clusters, once I had confirmed that the code was running on a 30x30 cluster, I left the programme to run overnight to obtain the graphs.
- Updated Lab Book to Include Equations list
 - As well as updating the Lab Book to reflect this week's work, I also created a page detailing the equations used in this project.
 - I waited until now to do this as I didn't know exactly what I would include / exclude due to the report constraints.
 - I have been keeping them stored on another platform until this point.

7.0.0.2 Main Results:

- Fully working code for 3D, producing 5 relevant and functional graphs.
- Nixed in on the 2D graphs to focus them towards the report, adapting these to reflect the data for multiple temperatures.
- Updated Lab Book to reflect this week's work as well as the equations sheet.

8 Week 6 Overview

8.0.0.1 Main Activities:

- Adapted Code in 2D to aid running efficiency
 - Whilst the 2D code produced in week 5 was running, it was very slow to produce the 300x300 cluster, running with 20,000 steps.
 - I re-worked the code so that it would run in one block on Jupyter Notebook, as it seemed this could decrease run times.
 - I also added a section to the code to produce a plot for each temperature output, visually displaying
 - the cluster shape and largest cluster size.
- Continued with Report Writing
 - Further to my previous report writing done for this project, I started to focus on the theory, methods, results and references sections.
 - For the theory / methods section, I have used the equations compiled in week 5, along with an overview of the coding methods used.
 - For the referencing, I have used mybib.com to produce the citations, using Vancouver referring.
 - I have decided to include these in the report as plain references, rather than employing the specific latex functionality, as previously that has been very unsuccessful.
 - Once the code has compiled, I then began by inputting the figures into latex, ensuring that the sizing was uniform, orderly and optimal, from both readability and space-saving perspectives.
 - I then proceeded to write the captions for all of these graphs.
 - Given the limited space, I selected the cluster graphs that contained the most information and best assisted the narrative, and created a single plot to contain these 4 graphs.
- Finalised the Lab Book
 - Given this is the last week of experiments, I decided to finalise the lab book.

- I ensured that the weekly log was complete and up-to-date, that the equations page was functioning and that the main page included relevant and completed information.

8.0.0.2 Main Results:

- Adapted the 2D code to aid running efficiency, ensuring large cluster sizes could be sustainably achieved.
- Continued writing the report, focusing on the theory, methods, results and references sections.
- Updated Lab Book to reflect to ensure that the weekly log was complete and the main page was detailed, ready for submission.

9 Week 7 & 8 Overview

9.0.0.1 Main Activities:

- Worked on / Completed Report
 - I commenced writing the report, doing so in several drafts so that all information was considered before shortening the report to fit in to the 4 pages.
 - I decided to prioritise space to results / discussion, rather than introduction and theory.
 - I tried to recompile to produce Figure 2, to make the colours more visually pleasing. Unfortunately, this was not possible due to Python Kernel failure.
 - I carefully selected the best / most influential equations, which would contribute most to the narrative of the report.
 - I ensured that references were present in the report and that citations were made accordingly.
 - I edited the report thoroughly.
- Finalised the Lab Book
 - I continued with the Lab Book log for weeks 7 and 8.
 - I ensured that the weekly log was complete and up-to-date, that the equations page was functioning and that the main page included relevant and completed information.

9.0.0.2 Main Results:

- Worked on and Completed the report, ensuring that the most important information was included and that the most meaningful equations were present, as well as that references were cited accurately.
- Finalised the Lab Book to ensure that it was all in order for submission.

10 Equations

10.1 Inverse temperature

$$\beta = \frac{1}{k_B T}$$

$$\beta = \frac{1}{T}$$

10.2 Spin variables

$$s_i \in \{+1, -1\}$$

10.3 Total number of sites

$$N = L^d$$

$$N = L^2$$

10.4 Ising Hamiltonian

$$H = -J \sum_{\langle i, j \rangle} s_i s_j$$

$$H = - \sum_{\langle i, j \rangle} s_i s_j$$

10.5 Energy change for a trial spin flip

$$\Delta E = 2s \sum_{\text{nn}} s_j$$

10.6 Metropolis acceptance rule

$$P_{\text{acc}} = e^{-\beta \Delta E}$$

10.7 Periodic boundary condition

$$x \equiv x + L$$

10.8 Magnetisation

$$M = \sum_i s_i$$

$$m = \frac{M}{N}$$

10.9 Energy of a configuration

$$E = - \sum_{\langle i,j \rangle} s_i s_j$$

10.10 Cluster size

$$n_k = \#\{i \in k\}$$

10.11 Cluster number density

$$z(n) = \frac{c_n}{N}$$

10.12 Mean cluster number density

$$\bar{z}(n) = \frac{1}{M} \sum_{m=1}^M z_m(n)$$

10.13 Standard error of the mean

$$\text{SEM}[z(n)] = \frac{\sigma[z(n)]}{\sqrt{M}}$$

10.14 Power-law cluster size distribution

$$z(n) \propto n^m$$

$$\log z(n) = m \log n + b$$

10.15 Radius of gyration squared

$$R_g^2 = \frac{1}{n} \sum_{i=1}^n |r_i - r_{cm}|^2$$

10.16 Circular mean mapping

$$\theta = \frac{2\pi x}{L}$$

$$z = \langle e^{i\theta} \rangle$$

10.17 Cluster extent

$$\text{extent} = x_{\max} - x_{\min} + 1$$

10.18 Spanning criterion

$$\text{extent} > 0.8L$$

10.19 Cyclomatic number

$$c = \ell - n + 1$$

10.20 Compactness measure

$$A = \frac{c}{(n-1)^2}$$

10.21 Largest-cluster fraction

$$f_{\max}(t) = \frac{n_{\max}(t)}{N}$$

10.22 Autocorrelation function

$$A(\tau) = \frac{\langle (x_t - \bar{x})(x_{t+\tau} - \bar{x}) \rangle}{\langle (x_t - \bar{x})^2 \rangle}$$

10.23 Decorrelation time

$$\tau_c = \min \{ \tau : A(\tau) < e^{-1} \}$$

10.24 Logarithmic binning

$$n_b \in \text{log-spaced bins}$$